

"STUDIES ON THE DRY ROOT-ROT OF CHICKPEA"

BY

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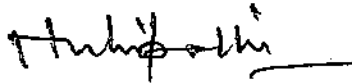
1985

**DEDICATED
TO
MY
GRAND MOTHER**

CERTIFICATE I

This is to certify that this thesis entitled "Studies on the dry root-rot of chickpea", submitted for the degree of M.Sc., in the subject of Plant Pathology, of the Haryana Agricultural University, is a bonafide research work carried out by Shri Ranbir Singh Taya under my supervision and that no part of this thesis has been submitted for any other degree.

The assistance and help received during the course of investigation have been fully acknowledged.


MAJOR ADVISOR


CO-MAJOR ADVISOR

CERTIFICATE II

This is to certify that the thesis entitled
"Studies on the dry root-rot of chickpea" submitted
by Shri Ranbir Singh Taya to the Haryana Agricultural
University in partial fulfilment of the requirements
for the degree of M.Sc., in the subject of Plant
Pathology, has been approved by the Students' Advisory
Committee after an oral examination on the same, in
collaboration with an External Examiner.

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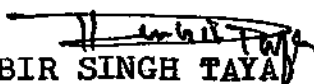
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INTRODUCTION

INTRODUCTION

Chickpea (Cicer arietinum L.) is an important high proteinaceous rabi pulse crop grown all over the world. It was cultivated over 7308.4 thousand hectares with a total production of 4754.9 thousand tonnes in India and in Haryana, this crop was cultivated over 647 thousand hectares with a total production of 315 thousand tons during 1983-84. Haryana state contributes 18 per cent of the total production in the country.

Chickpea suffers from various insect-pest and diseases as well as natural calamities. However, diseases take heavy toll as compared to other menaces. Among the various bacterial, fungal, viral and nematode diseases, dry root-rot is a major disease which destroys the crop maximum particularly in Haryana, where chickpea is cultivated as 'Barani' crop. This disease is characterized by the yellowing and drooping of petioles and leaflets from the lower portion of the plants. Affected plants scattered in patches and show stunted growth. Light brown patches are found on the root. The lesions start from tips, but, it can also be seen in between the root zones. Dry root-rot appears at seedlings as well as flowering stage of the crop. Rhizoctonia bataticola has been reported the cause of this disease.

R. bataticola is an omnipresent notorious pathogen which survives in the soil. It has, therefore, to interact with

different other soil micro organisms including soil borne chickpea pathogens. Under such circumstances, the damage to the crop is expected more. Being a soil inhabitant, R. bataticola is bound to be influenced by various factors like soil texture, moisture, temperature and nutritional status. However, these factors have not been correlated in relation to dry root-rot development of chickpea.

Several chemicals have been tried to control diseases caused by R. bataticola but no work has been done on the control of dry root-rot of chickpea caused by R. bataticola. After considering the meagre work done on the various aspect of such devastating disease, attempt has been made to study the problem on the following lines:

1. Survey for the incidence of dry root-rot of chickpea in farmer's fields.
2. Isolation and purification of the pathogen and pathogenicity test.
3. Studies on the interaction between R. bataticola and F. oxysporum f.sp. ciceri.
4. Effect of nitrogen & phosphorus doses, soil type on dry root-rot development.
5. Screening of fungicides against the pathogen under laboratory conditions.
6. Studies on the control of dry root-rot by seed treatment and soil drenching methods.

7. Studies on the effect of soil type on disease controlling potential of fungicides,
8. Studies on the effect of nitrogen and phosphorus doses on the disease controlling potential of the fungicides.

It is hoped that the results of these studies will be helpful to the science of Plant Pathology as well as extension workers and farmers to understand and control dry root-rot of chickpea.

REVIEW OF LITERATURE

REVIEW OF LITERATURE

Chickpea (Cicer arietinum L.) suffers from vagaries of a number of bacterial, fungal and viral diseases. Recently Nene (1984) named about 32 diseases which are responsible for the reduction in yield and quality of chickpea. Among the root diseases (dry root-rot, wet root-rot, Trichoderma root-rot, black root-rot, Pythium rot/seed rot, Fusarium wilt, Verticillium wilt) dry root-rot takes heavy toll every year specially in tropical countries. However, dry root-rot caused by R. bataticola (Taub.) Butl. has been found severe under Northern Indian condition. Nene et al. (1978) have characterized dry root-rot affected plants showing yellowing and drooping of petioles and leaflets from the top. Tripathi and Sharma (1983) have noted the dry root-rot at seedling stage from Southern part of Haryana. They have observed that light brown patches on the roots or the lateral rootlets, which showed light brown discolouration at the initial stage. The lesions later on, girdled the infected roots and rootlets, with the result plant dried-off. It appears at seedlings as well as flowering stage when the temperature becomes congenial for the growth of the fungus.

R. bataticola (Taub.) Butl. has been found associated with the causation of dry root-rot of many crops. Ghaffer and Zentmyer (1968) reported that this fungus attacks nearly 300 host plants in different parts of the world and more in the

tropical regions which mainly cause root-rot and stem rot. Ashby (1927) reported pycnidial stage of R. bataticola as Macrophomina phaseolina (Maub.) Ashby on bean tissue. He had noted that initial growth of this fungus was white but with age, it turn dark brown or black with abundant black sclerotia.

Effect of media on growth of R. bataticola:

Literature revealed that different isolates of R. bataticola require different types of media for growth and sclerotial formation. Uppal/^{et al.}(1936) while working with sorghum isolate of R. bataticola found that Meyer's solution and Richard's solution media good for linear growth as well as mycelial weight and Quaker and Richard's for sclerotial production. Soyabean isolates of R. bataticola flourished well on potato dextrose agar medium (Dhingra and Sinclair, 1972, 1973; Mukhopadhyaya and Nandi, 1975). Jain et al. (1972) found that urd bean leaf extract agar medium was better for linear growth. Richard's solution and Richard's agar medium were ideal for dry mycelial weight as well as linear growth than other media tested (Chattopadhyay and Bhattacharya, 1966; Ghosh and Sen, 1973; Shanmugam and Govindaswamy, 1973) while working with the groundnut and jute isolates. Rana (1980) found that linear growth as well as mycelial weight was maximum on Sabouraud broth and Richard's medium. Grover and Sakhuja (1981); Singh and Chohan (1982) found Czapek's agar medium ideal for growth of R. bataticola isolates of mungbean and muskmelon respectively.

Effect of media on the growth of *F. oxysporum*

To see the growth behaviour of *F. oxysporum* on different media, Buldeo and Rane (1978) found that growth of (sesamum isolate) *F. oxysporum* f.sp. *sesami* was maximum on Richard's solution followed by Czapek's medium. On the other hand linear growth and sclerotial formation was maximum on potato dextrose agar (PDA) medium. Eswara and Chaudhary (1985) tested different media and found that the growth of *Fusarium* *Udum* in liquid media was superior on Richard's solution followed by Czapek's and Brown's solution.

Effect of temperature on growth of *R. bataticola*

Different workers while working on the temperature requirement for growth and development of different isolates of *R. bataticola*, have noted that a temperature of 30 to 35°C was optimum for growth and sclerotial formation (Jain and Kulkarni, 1965; Ahmed and Ahmed, 1969; Grover and Sakhuja, 1981; Singh and Mehrotra, 1982). While some workers reported that optimum temperature for growth of *R. bataticola* was 30°C (Dhingra and Sinclair, 1972; Singh *et al.*, 1973^a; Shokes *et al.*, 1977 and Rana, 1980). Jain *et al.* (1972) reported that optimum range of temperature for growth of *R. bataticola* was in between 25 to 30°C. Mitra (1931) and Agarwal *et al.* (1973) found that *R. bataticola* caused maximum damage to crop in the range of 33 to 40°C temperature.

Effect of temperature on growth of *F. oxysporum*:

While working with the different isolates of *F. oxysporum*, Szatala (1962); Micoso and Ilays (1977) and Shukla and Bhargwa (1976) reported that fungus grew in culture and caused maximum damage to crop in between 20 to 30°C temperature, but Chohan (1963) studied the wilt incidence of gram and reported maximum damage to crop in pot experiment was at 25°C. However, disease incidence at lower and higher temperature was recorded low.

Interaction among pathogens:

Kerr (1963) while studying root-rot of pea observed that infection by *F. oxysporum* f.sp. *pisi* was enhanced when inoculum of *Pythium ultimum* was mixed with inoculum of *F. oxysporum* f.sp. *pisi*. A synergistic effect of mixed inocula of *F. solani* f.sp. *pisi*, *R. bataticola*, *Pythium ultimum* and *F. oxysporum* f.sp. *ciceri* was noticed by Westerlund et al. (1974) which caused wilt complex in chickpea. Pieczarka and Abawi (1978) noted mixture of *F. solani* f.sp. *phaseoli*; *Pythium ultimum* and *R. solani* applied as mixed inocula reduced the shoot length and per cent dry weight of bean plant, showing synergistic effect.

Effect of soil texture on root-rot diseases:

Philip et al. (1969) reported that *R. bataticola* caused more disease when sand percentage increased in the normal soil. Herr (1976) while studying sugarbeet root-rot found that inoculum of *R. solani* increased when crop was grown in sandy-loam

soil as compared to silt clay soil during growing season. Chattopadhyay and Mustafee (1977) working with different isolates of R. bataticola found that population in sterilized sandy loam soil increased markedly at pH 5 and 6, but at pH 8, there was a readily decline in population. Malhan (1980) reported that maximum disease was caused by R. bataticola on cowpea in sandy loam soil than silt loam or clay soil. Disease decreased as the clay or silt particles increased in normal soil.

Effect of nitrogeneous and phasphatic fertilizer on root-rot diseases:

Nitrogen is the most important single nutrient that helps in rapid growth of field crops and applied frequently. Thakurji (1974) reported that application of nitrogen alone increased incidence of stem rot and root-rot of jute caused by R. bataticola. As he increased the nitrogen doses, disease incidence increased significantly. Similarly, Ghosh and Basak (1965) found that ammonical nitrogen increased the disease and nitrate nitrogen reduced the damping-off and black-rot of sugarbeet. Whereas root-rot in sugarbeet (S. rolfsii) was controlled by application of 160 kg N/ha as calcium ammonium nitrate (Thakur and Mukhopadhyaya, 1972).

Phosphorus plays an important role in disease control. Thayer and William (1960) reported that an increase in concentration of phosphorus in gravel culture reduced the development of both stalk and root-rot of corn (M. phaseolina).

Gaur and Vaidya (1983) found that a progressive reduction in root-rot of gram (R. solani) occurred with increasing level of P_2O_5 . Incidence was least when 40 kg P_2O_5 /ha combined with 3 kg zinc sulphate/ha.

Sivaparkasam et al. (1975) found that root-rot of Sunflower (M. phaseolina) increased as the doses of nitrogen alone increased but decreased as the doses of potassium and phosphorus increased, whereas, Ramasami and Shanmugam (1976) reported that deficiency of potassium, phosphorus and sulphur led to increase of the root-rot of sunflower.

Screening of Fungicides:

In in vitro:

A large number of fungicides have been screened against isolates of R. bataticola by poisoned food technique. Amongst the protectants quintozone, Captan, Captafol, ceresan dry were found effective as growth inhibitor under laboratory condition against various isolates of R. bataticola (Masih ^{et.al.,} 1978; Jain and Khare, 1972; Malhan, 1980; Hooda and Grover, 1983; Rana and Tripathi, 1983). The doses varied according to isolates. Similarly carbendazim, benomyl, thiophanate-methyl, carboxin, among the systemic fungicides were found effective in in vitro studies as growth inhibitory (Jain and Khare, 1972; Rath and Mishra, 1978; Malhan, 1980; Goyal and Mehrotra, 1981; Kotwal, et al., 1981; Taneja and Grover, 1982; Hooda and Grover, 1983; Rana and Tripathi, 1983 and Dharamvir, 1984).

Seed Treatment:

Seed treatment has been found effective for the control of root-rot disease. Lewin and Nataranjan (1971) reported that seed treatment with aresan, ceresan, captan and thiram against R. bataticola isolated from groundnut established higher seed germination and have minimized post emergence root-rot incidence to a considerable extent. Seed treatment with agrosan GN was effective in controlling seed borne infection and collar rot of double bean (Petker et al., 1977). Similarly root-rot incidence of chick-pea was reduced by seed treatment with thiram + either benomyl or captan (Cotter, 1977). Carbamate fungicides has been reported to be effective in controlling disease caused by R. bataticola on cowpea and sesamum (Sinha and Khare, 1977; Tripathi et al., 1977) respectively, however, thiram was not found effective against chickpea root-rot (Verma and Vyas, 1977).

Benzimidazole group of fungicides gave promising results in controlling the various diseases caused by R. bataticola. Verma and Vyas (1977) found that S. rolfsii and R. bataticola were controlled on gram by carboxin and F. oxysporum f.sp. ciceri by carbendazim. Seed treatment with carbendazim alone or in combination with other fungicides controlled the various isolate of R. bataticola (Chohan et al., 1979; Malhan, 1980; Mishra, 1980; Basman and Prasad, 1981; Chatopadhyaya et al., 1981; Goyal and Mehrotra, 1981; Shukla & Singh, 1973; Vyas et al., 1981; Taneja and Grover, 1982; Hooda and Grover, 1983; Rana and Tripathi, 1983 and Dharamvir, 1984). Grover and Chopra (1977)

found that carboxin and oxycarboxin have chemotherapeutic activity in okra seedlings against R. bataticola while treated seed were sown.

Soil drench:

Soil drenching with vapam and formaldehyde completely controlled the M. phaseolina within 5 days and F. solani within 10 days (Mustafee and Chatopadhyay, 1981). Soil drenching with quintozone, captan, ferbam, thiram and plantvax gave good control of sugarbeet root-rot and muskmelon root-rot (Singh et al., 1973; and Suhag and Duhan, 1983) respectively. Mathur and Singh (1973) reported that soil treatment with quintozone at the rate of 50 lb/acre ten days before sowing control root-rot in cotton (R. bataticola). Oladizan and Okusanya (1980) found that soil drench with thiram, thiram + BHC, captafol gave good results in controlling pre and post emergence damping-off of cowpea (R. bataticola). Soil drenching with carbendazim or benlate have been found effective to control wilt and stem rot of carnation and root-rot of muskmelon (Shehata et al., 1982 and Suhag and Duhan, 1983) respectively. Petker/^{et al.}(1977) while studying with groundnut isolate of R. bataticola found that soil treatment with carboxin gave better control than seed treatment with carbendazim and seed treatment and soil drench also gave good control with carbendazim in sesamum (Tripathi et al., 1977).

Grover and Chopra (1977) and Hooda and Grover (1982) have reported superiority of carboxin for the control of pre and post emergence root-rot of okra and cowpea, respectively.

Suhag and Rana (1984) reported that soil drenching with carbendazim followed by a second drench with captafol or thiram gave maximum protection against Rhizoctonia rot of onion.

Effect of soil types on the efficacy of fungicides:

In in vitro studies, Rushdi and Jeffers (1956) reported that percentage inhibition of growth of R. solani by dithane D-10 and actidione was significantly reduced as the amount of clay and silt content increased in soil. Ilyas et al. (1976) found that benomyl, thiophanate-methyl, thiabendazole, triforine, thiram and captan controlled less disease of cowpea caused by R. bataticola in silty loam soil with high organic matter than sandy loam soil with low organic matter. Malhan (1980) studied the control of root-rot to cowpea (R. bataticola) with thiophanate-methyl, difolatan, captan, bavistin and reported that minimum disease was recorded in sandy loam soil and maximum in clay loam soil. Disease controlling efficacy of fungicides increased with addition of sand particles in soil and decreased when silt and clay particles added.

Effect of nitrogen and phosphorus on disease controlling efficacy of fungicides:

Yadav (1980) found that application of nitrogen through urea led to reduction in disease controlling potential of the MEMC, quintozone and carbendazim, but nitrate form of nitrogen reduced the efficacy of MEMC and quintozone more than carbendazim, while phosphorus improved the efficacy of carbendazim, with the result of increasing disease controlling ability.

Malhan (1980) reported that disease controlling ability of carbendazim and thiophanate-methyl on cowpea (R. bataticola) was significantly reduced in the presence of potassium or in combination with phosphorus. In case of difolaton, phosphorus alone or in combination with nitrogen in soil led to reduction in disease control.

Shyam Sunder (1982) observed that disease controlling ability of MEMC, quintozone, calixin and carbendazim was severely reduced with N+P+K fertilizer when applied as combination against R. solani.

MATERIALS AND METHODS

MATERIALS AND METHODS

1.1 Survey on the dry root-rot of chickpea:

Survey for the incidence of dry root-rot of chickpea was conducted in different farmers fields of Hisar and Kurukshetra districts. The disease incidences were recorded at two stages i.e. seedling stage (preflowering) and at maturity stage (Pod formation). Five, randomly selected two meter square area (containing roughly 200 plants) were selected from a field. Total plants as well as infected plants were counted and the per cent disease incidence was calculated. Two to ten fields from one village were surveyed for this purpose. Observations were recorded from each field at a location and thus seven locations from Hisar and two locations from Kurukshetra district were surveyed. Data on disease incidence are presented in Table-1.

1.2 Isolation, purification and maintenance of the organisms:

Ten randomly selected plants from each sample were brought in the laboratory for further studies. Isolations were made from diseased portion of the root of chickpea seedling on Potato dextrose agar (PDA) medium by the method described by Riker and Riker (1936). Cultures were purified by hyphal tip method four to five times and thus the organism(s) tentatively identified and mentioned in Table-1. Stock cultures of the pathogens were maintained on Potato dextrose agar slants in refrigerator at 8°C and renewed after every four weeks.

1.3 Pathogenicity test:

Inocula of R. bataticola (Taub.) Butl. and F. oxysporum f.sp. ciceri (Pedw.) Matvo and Sato raised on boiled gram (100 gm of gram were soaked in 150 ml of water for twelve hours and autoclaved at 20 lb pressure per square inch for 30 minutes and incubated at $28\pm 2^{\circ}\text{C}$ in B.O.D. incubator for seven days. Inocula of R. bataticola and F. oxysporum f.sp. ciceri were mixed in sterilized sandy loam soil (soil was autoclaved at 20 lb pressure per square inch for one hour after a gap of 24 hours) in different ratio. This inoculated soil was put in 15 cm diameter earthen pots for two days. Fifteen seeds of chickpea (cv. C-235) were sown in each pot in three replications set. Pots were irrigated regularly at an interval of three days. Diseased plants along with healthy plants were counted after 25 days of sowing. Data on disease incidence were calculated on the basis of healthy and diseased seedlings.

2.1 Effect of different media on growth of R. bataticola and F. oxysporum f.sp. ciceri:

Eight different natural and synthetic media (Appendix-I) were selected to study the growth behaviour of R. bataticola and F. oxysporum f.sp. ciceri. Initial pH of all the media was adjusted to 6.5 before autoclaving. To find out the growth in liquid media, agar-agar was not added. Required amounts of constituents were dissolved in sterilized water and 40 ml of these media were put in 150 ml of Erlenmeyer flask and were sterilized at 15 lb pressure per square inch for 20 minutes in

autoclave. Four replications of each treatment were maintained. Each flask was inoculated with a mycelial disc of 5 mm diameter from the periphery of actively growing cultures in Petriplates. (20 ml, sterilized PDA in 90 mm diameter Petriplate). To find out the growth behaviour on solid media agar-agar was added as per requirement. After autoclaving, media were poured in 90 mm diameter sterilized Petriplates (at 180°C for 2 hours in an oven) in four replicated sets of each treatment. These Petriplates were inoculated with the same size mycelial disc, as mentioned earlier. Inoculated flasks and Petriplates were incubated at 28±2°C for seven days and sixty hours respectively in B.O.D. incubator. The fungal growths on solid media were measured with the help of scale after interval of 12 hours. Mycelial mat was filtered through pre-weighed Whatman filter paper No. 1 and dried at 60°C for 24 hours. Weight of mycelial mats were obtained from each flask by detecting the weight of filter papers.

2.2 Effect of temperature on growth of *R. bataticola* and *F. oxysporum* f.sp. *ciceri*:

To find out the optimum temperature for growth and development, *R. bataticola* and *F. oxysporum* f.sp. *ciceri* were grown in Richard's nutrient and without agar in Petriplates and flasks respectively. Inoculated flasks and Petriplates were incubated at different temperature ranging from 10 to 40°C in three replicated sets. Data on linear and dry mycelial growth, obtained by the methods described in previous experiment are presented in Fig. 3 and 4.

3.1 Effect of *F. oxysporum* f.sp. *ciceri* on growth of *R. bataticola* in in-vitro:

To find out the synergistic and/or antagonistic effect of *F. oxysporum* f.sp. *ciceri* on the growth of *R. bataticola*, four days old cultures of both the pathogens were grown on Richard's agar medium in Petriplates. The bits of both the pathogens were put in three different fashions viz. I) Bits of both the pathogens were put together in the centre of the Petriplates. II) Bit of *R. bataticola* of same size was put in the centre while four bits of *F. oxysporum* f.sp. *ciceri* were put on the periphery of Petriplates at equal distances. III) Bit of *F. oxysporum* f.sp. *ciceri* was put in the centre while four bits of *R. bataticola* were put on the periphery of Petriplates at equal distances. After inoculation these Petriplates were incubated at $28 \pm 2^{\circ}\text{C}$ in B.O.D. incubator. Radial growth was measured after 48 hours.

3.2 Effect of *F. oxysporum* f.sp. *ciceri* on growth of *R. bataticola* in in-vivo studies:

To find out the effect of *F. oxysporum* f.sp. *ciceri* on *R. bataticola* in in-vivo following two experiments were conducted:

3.2.1 Technique proposed by Wansley and McKeen (1962) were modified to study the interaction effect between *R. bataticola* and *F. oxysporum* f.sp. *ciceri*. Seven days old mycelial culture of both the pathogens were homogenized in homogenizer for 30 seconds at the rate of one gram fresh mycelial mat weight in 100 ml of distilled sterilized water. Mycelial suspension

of both the pathogens were further diluted and mixed with each other in different ratios. Five days old gram seedlings raised in pure sterilized sand were taken and washed thoroughly with double distilled water in such a way that no injury were caused on roots and rootlets. Mycelial suspension at the rate of 40 ml per test tube were taken and one seedlings was rolled with sterilized absorbant cotton in such a way that roots were dipped in suspension while shoot remained on the upper side of the test tube and thus cotton served as plug. Nine treatment along with three control (I) R. bataticola, II) F. oxysporum f.sp. ciceri, III) Pure double distilled sterilized water) replicated ten times, were kept in light fitted B.O.D. incubator at $28 \pm 2^{\circ}\text{C}$ for ten days. Symptoms started appearing after 40 hours. Final reading on disease incidence, shoot and root length were recorded after 72 hours and 96 hours respectively.

3.2.2 Pot culture test:

Inocula of both the pathogens, raised on boiled gram were mixed with sterilized sandy loam soil at the rate of 5 per cent on weight basis. Inoculated soils were kept in 15 cm diameter earthen pots separately as well as in mixture of both pathogens (50:50 ratio of each pathogen). Four replications of each treatment along with control (uninoculated sterilized soil) were maintained. Data on disease incidence, shoot length and root length were recorded after 25 days of sowing.

4.1 Effect of soil type on disease incidence:

Five different types of soil having different textural characteristics (Appendix-II) were collected from different locations and got analysed from department of Soil, Haryana Agricultural University, Hisar. These soils were sterilized and inoculated with 5 per cent (W/W) boiled gram inocula. These inoculated soils were kept in 15 cm diameter earthen pots in five replications along with control (same type of soil uninoculated). Fifteen seeds were sown in each pot. Per cent disease incidence was recorded after 25 days.

4.2 Effect of nitrogen and phosphorus on the development of dry root-rot of chickpea:

Chickpea required 15 kg of nitrogen and 40 kg phosphorus per hectare as per University recommendation. Therefore, an experiment was conducted to find out the effect of one higher and one lower level of nitrogen and phosphorus as well as without fertilizer respectively on the incidence of dry root-rot of chickpea. In this experiment nitrogen and phosphorus were supplied by urea (46 per cent nitrogen) and single super phosphate (16 per cent phosphorus) respectively. The required amount of nitrogen and phosphorus were calculated on soil weight basis. Fertilizers applied as N_0 - 0 kg N/ha; N_1 - 10 kg N/ha; N_2 - 15 kg N/ha; N_3 - 20 kg N/ha; P_0 - 0 kg P_2O_5 /ha; P_1 - 20 kg P_2O_5 /ha; P_2 - 40 kg P_2O_5 /ha; P_3 - 60 kg P_2O_5 /ha. Sandy loam soil analysed for N.P.K. level (.37 kg N/ha; 4 kg P_2O_5 /ha;

230 kg K/ha) inoculated with 2.5 per cent (W/W) R. bataticola inoculum raised on boiled gram and mixed with different levels of fertilizers. Inoculated soil was kept in 30 cm diameter earthen pots in polythene bags alongwith control (no fertilizer) in three replications. Twenty seeds of gram (cv. C-235) were sown in each pot and pots were irrigated regularly. Data on disease incidence ~~were~~ recorded after 30 days of sowing.

5.0 Screening of fungicides:

5.1 In in-vitro:

Fungicides belonging to different groups of chemical were screened (formulation basis) by poisoned food technique as described by Schimtz (1930); Palmiter and Keith (1937) and Carpenter (1942) using Richard's agar medium. Growth inhibition by different fungicides at different doses were calculated. Toxicity index and ED_{50} were determined by the method described by Rich and Horsfall (1961) and Horsfall and Luken (1965). Detail of fungicides used are given in Appendix-III. Toxicity index in the present study is based on the total inhibition of pathogens in all the concentrations.

5.2 Seed treatment method:

Eight fungicides (farmulation basis) were applied to treat the seed at the rate of 0.2 per cent (W/W) before sowing. Required amount of fungicides and seeds (200 mg of fungicides/ 100 gm of seed) were put in 250 ml Erlenmeyer flask and shaken to coating the surface of the seed with fungicides thoroughly.

Fifteen treat-ed seed were sown in inoculated sterlized (5 per cent W/W) soil in 15 cm diamter earthen pot. Four replications of each treatment along with control (untreated seed) were maintained. Per cent disease incidence and final stand were determined.

5.3 Pre-sowing drenching of soil:

Inoculated sandy loam soil were drenched with eight fungicides (formulation basis) at the rate of 0.2 per cent(W/V) except control. Fifteen untreated seed were sown in each 15 cm diameter earthen pot. Pots were irrigated regularly. Per cent disease incidence and final stand were calculated after 25 days of sowing.

5.4 Seed treatment and soil drenching:

Treated seeds with eight fungicides (formulation basis) were sown in inoculated sandy loam (5 per cent W/W) and drenched after germination of seedlings with the same dose of fungicides. Pots were irrigated regularly. Data on disease incidence and final stand were calculated after 25 days of sowing.

6.1 Effect of soil types on the efficacy of selective fungicides:

6.1.1 Seed treatment:

Seed treated with carbendazim and carbendazim + thiram (1:1 ratio) at the rate of 0.2 per cent W/W were sown in inoculated different four types of soils. Further detailed of this method is given in previous experiment.

6.1.2 Soil drenching:

Inoculated different types of soil were kept in pot for 5 days, so that the inoculum may stablize. These soil were

drenched with carbendazim and carbendazim + thiram two days before sowing. Untreated fifteen seeds were sown in each pot. Data on disease incidence and final stand were recorded after 25 days of sowing.

6.2 Effect of nitrogen and phosphorus on the efficacy of selective fungicides:

6.2.1 Seed treatment:

Different levels of nitrogen and phosphorus in inoculated soil were maintained in 30 cm diameter earthen pots as described earlier. Treated seeds with carbendazim and carbendazim + thiram (50:50 ratio) were sown in each pot with four replications. Data on disease incidence and final stand were calculated 30 days after sowing.

6.2.2 Soil drenching:

Different levels of nitrogen and phosphorus in inoculated soil were maintained as described earlier and kept for 5 days for stabilizing of the pathogen. The pots were drenched with carbendazim and carbendazim + thiram solutions at the rate of 0.2 per cent (W/V) (500 ml suspension/2 kg of soil).having 4 replications of each treatment. Data on disease incidence and final stand were calculated 30 days after sowing.

Miscellaneous:

1. The seeds of chickpea (cv. C-235) were obtained from Nucleus Seed Production Section, Haryana Agricultural University, Hisar.

2. In most of the cases the numbers of replications were three or four and suitable control was maintained.
3. Most of the data were analysed statistically by using analysis of variance to determine significance.
4. The chemicals used were from BDH analytic reagent and glassware was 'Corning' brand.
5. Distilled water was used for most of the studies.

RESULTS

RESULTS

1.1 Survey of dry root-rot of chickpea:

To find out the incidence of dry root-rot of chickpea, thirty nine fields from nine different locations were surveyed during rabi 1983-84 and average disease incidence of each field were recorded. Data presented in Table-1 indicates that 35 per cent dry root-rot was noticed in Gangwa followed by 31 per cent Rawalwas (Hisar).

The lowest 12.5 per cent dry root-rot was recorded from fields of Kaul (Kurukshetra). Dry root-rot was also recorded in HAU Breeder Seed Production Area where disease incidences varied from 14 to 18 per cent. In general, it is clear from Table-1 that dry root-rot varied from 12.5 to 35 per cent.

1.2 Isolation and purification of organisms:

Samples collected from the fields were plated on the media to find out the involvement of the casual organism(s). Data presented in Table-1 indicates that most of the samples (68 to 90 per cent) were infected by R. bataticola which is characterized as having spreading type growth which was white at the beginning and turned black with age. Smooth, round to oval black sclerotia, measuring 70.0 to 117.0 x 72.0 to 108.0 μ in size were produced in large numbers. The lateral branches were coming out often at right angle with the constriction at the point of origin of lateral branches. Pycnidia were not produced in culture. These characteristics resembled with R. bataticola and hence identified as R. bataticola (Taub.) Butl. In few samples, F. oxysporum f.sp.

Table 1: Prevalence of dry root-rot of chickpea (R. bataticola) at different locations.

Sr. No.	Locations	Total fields observed	Percent disease incidence	Organism(s) isolated (percent)		
				<u>R. bataticola</u>	<u>F. oxysporum f.sp. ciceri</u>	Mixture of both
1.	Director Farm H.A.U. Hissar	5	18.0	90	-	10
2.	N.S.P. H.A.U. Hissar	10	14.0	85	6	9
3.	Village Gangwa	7	35.0	76	4	20
4.	Village Kurdi	5	24.4	68	2	30
5.	Village Rawalwas	3	31.0	72	1	27
6.	Village Muklan	2	27.0	80	-	20
7.	Village Pali	3	24.0	75	-	25
8.	Kaul H.A.U. Farm	2	12.5	68	10	22
9.	Village Sakra	2	16.0	76	8	16
Total		39				

ciceri was also isolated, but in these samples clearcut lesions were not found. F. oxysporum f.sp. ciceri has profusely branched white mycelium without any pigment. The micro conidia was cylindrical, straight, curved and oval (5-13.5 x 2-3.2 μ). Macro conidia general 3-5 septate, measuring 30-65 x 3.2 x 5.2 μ . Which was later on, confirmed by Common Wealth Mycological Institute, Kew (United Kingdom). In some samples, both R. bataticola and F. oxysporum f.sp. ciceri were found associated with diseased roots.

1.3 Pathogenicity test:

Inocula of R. bataticola and F. oxysporum f.sp. ciceri raised on boiled gram were mixed with sandy loam soil in the ratio of 2.5 : 97.5, 5:95, 10:90, 20:80 per cent. Per cent disease incidence was recorded after 25 days are given in Table-2.

In case of R. bataticola 55 per cent disease incidence was recorded at 2.5 per cent inoculum level and 83.3 per cent disease incidence at 5 per cent inoculum level. But at 10 and 20 per cent inoculum levels 100 per cent disease incidence was recorded. Yellowing of lower leaves were noted as visible symptoms. Light brown discolouration on the roots and rootlets were recorded in the initial stage of infection. There were no marked differences in root and shoot length as compared to control.

Table 2: Effect of inoculum concentration on dry root-rot of chickpea (R. bataticola)

Ratio of inoculum and soil (inoculum : soil)	Per cent disease incidence	Per cent final stand	Mean shoot length (mm)	Mean root length (mm)
2.5 : 97.5	55 ¹ (47.81) ²	45	116	114
5 : 95	83.3 (65.95)	16.70	118	112
10 : 90	100 (88.19)	-	115	110
20 : 80	100 (88.19)	-	115	112
Control (uninoculated)	-	100	118	114

C.D at 5 per cent (1.83)

1. Average of three replications,
2. Figures in parenthesis are angular transformed value.

In case of F. oxysporum f.sp. ciceri, no clearcut lesions were observed on the root and rootlets but plants were poor in growth and remained stunted. At 2.5 per cent inoculum level, the root and shoot length were 57 mm and 65 mm as compared to control (117 mm and 113 mm) respectively, but as inoculum level increased, the stunting also increased (Table-3).

Table 3: Effect of inoculum concentration on wilt of chickpea.

F. oxysporum f. sp. ciceri

Ratio of inoculum and soil (inoculum : soil)	Per cent disease incidence	Percent final stand	Mean shoot length (mm)	Mean root length (mm)
2.5 : 97.5	-	100	65	57
5 : 95	-	100	64	55
10 : 90	-	100	62	53
20 : 80	-	100	59	52
Control (uninoculated)	-	100	117	113

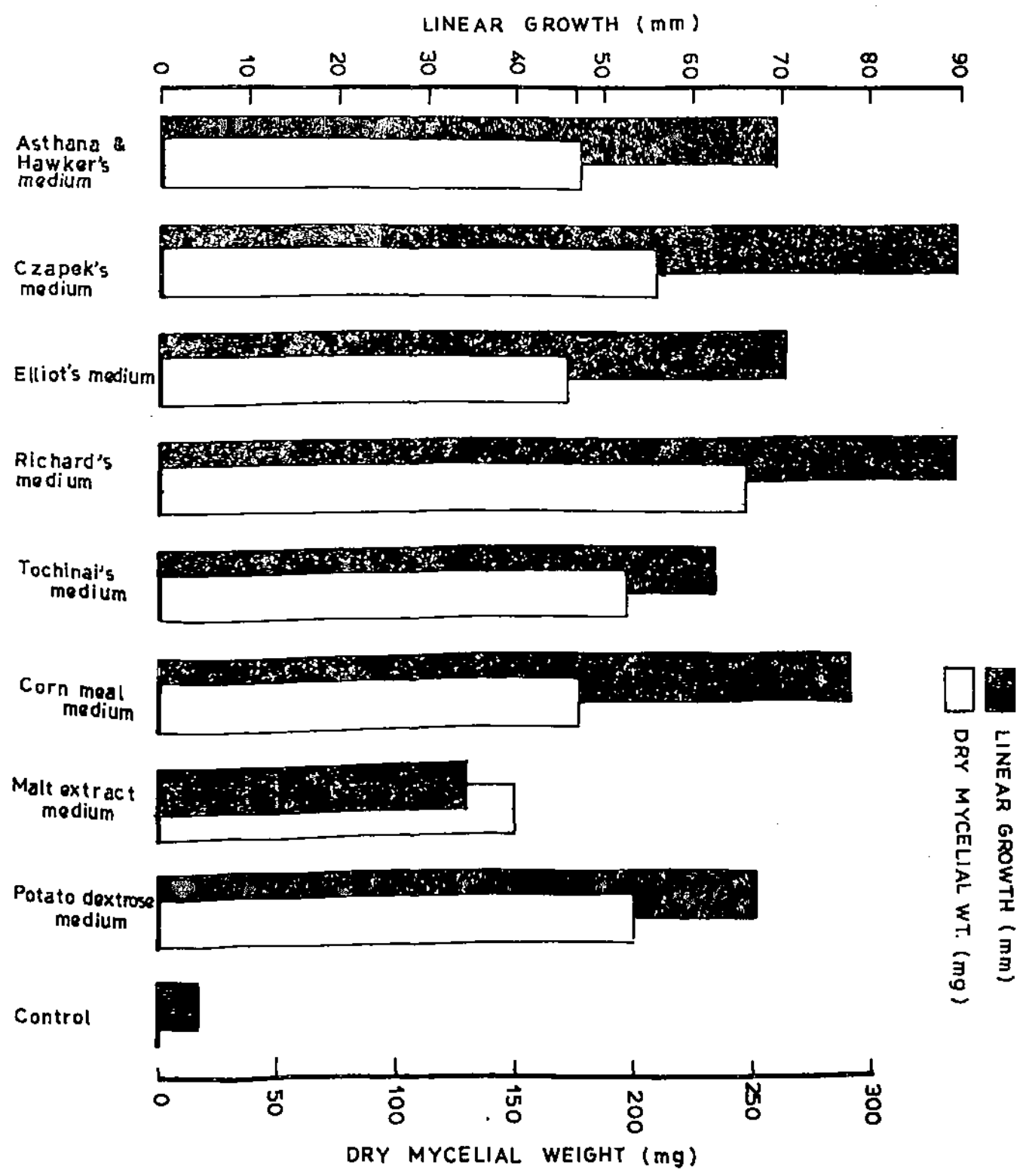
Average of three replications.

2.1 Effect of media on the growth of *R. bataticola*:

Asthana and Howker's, Czapek's, Elliot's, Richard's, Tochinnai's Malt Extract, Cornmeal and Potato dextrose media were selected to study the growth behaviour (linear as well as dry weight basis). Data presented in Fig.-I indicates that Czapek's and Richard's agar media supported maximum growth of 90 mm in 60 hours followed by Cornmeal and PDA media where growths were 78 and 72 mm respectively. Out of these media, the growth on Czapek's media was very thick. Sclerotial formation was more on Czapek's agar medium followed by Cornmeal agar medium. The sclerotial formation comparatively lesser on Richard's agar and PDA medium. The lowest growth was recorded on simple water agar medium.

The growth behaviour in liquid medium represented by dry mycelial weight basis were also observed and it was found that Richard's solution medium exhibited maximum (247 mg) dry mycelial

Fig 1. EFFECT OF MEDIA ON GROWTH OF *Rhizoctonia bataticola*



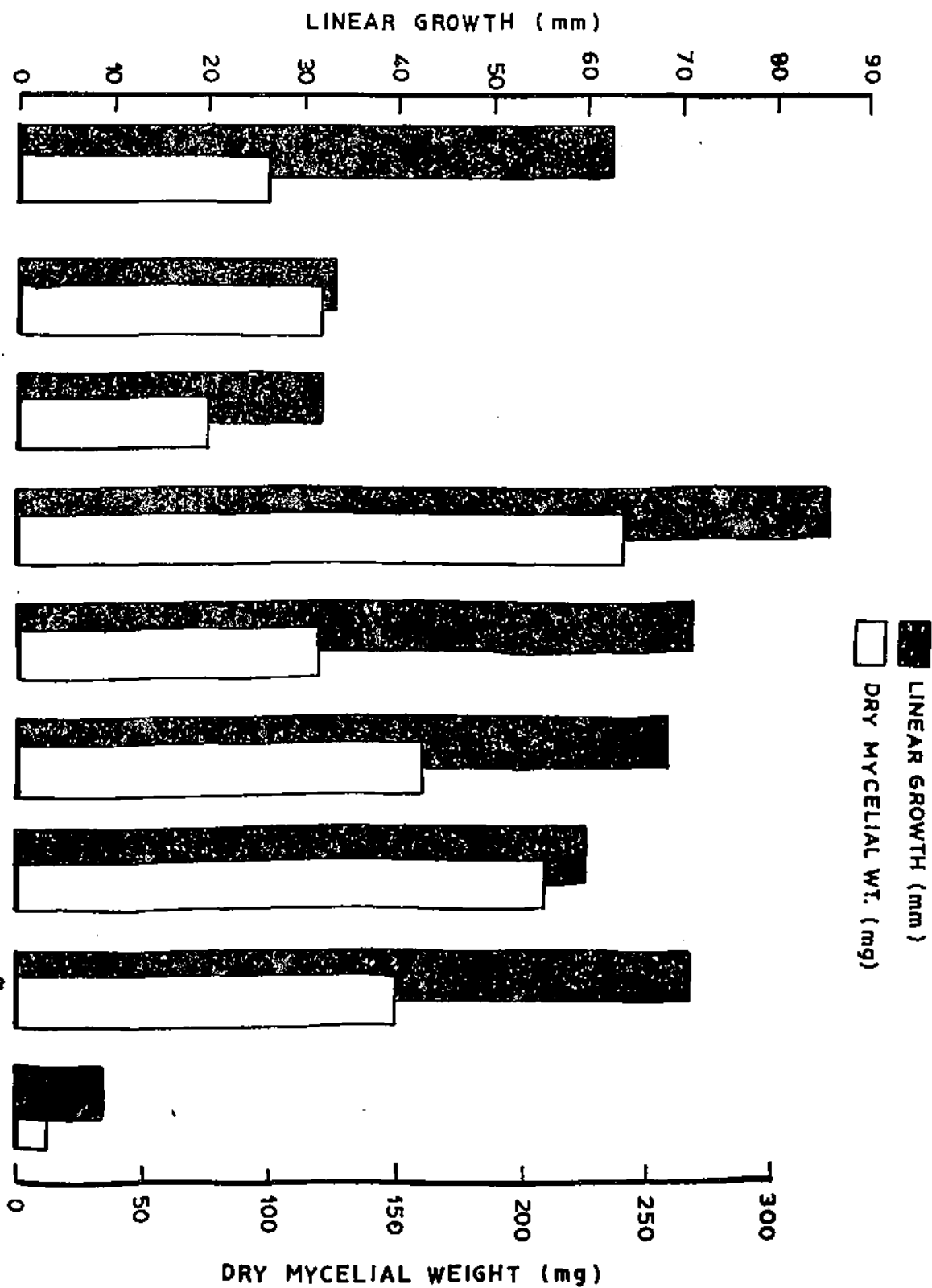


Fig. 2. EFFECT OF MEDIA ON THE GROWTH OF *Fusarium oxysporum* f.sp. *ciceri*

weight as compared to no growth in sterilized distilled water upto 7 days followed by Czapek's medium (218 mg). Lowest growth was recorded on malt extract medium. Among the natural media PDA exhibited maximum (198 mg) dry mycelial weight (Fig.-I).

Effect of media on the growth of *F. oxysporum* f.sp. *ciceri*:

Above mentioned eight media were selected to study the growth of *F. oxysporum* f.sp. *ciceri* on solid as well as on liquid media. Data presented in Fig.-2 indicates that Richard's agar medium gave maximum linear growth of 86 mm. The linear growths of PDA and Tochinai agar media were similar after sixty hours. The lowest growth was obtained on simple agar medium (12 mm). Among the synthetic media lowest growth was recorded on Elliot's medium (32 mm). Among the natural media, best growth was recorded on PDA medium (72 mm) followed by Cornmeal and Malt extract.

In case of liquid media the growth behaviour on different media were calculated on dry weight basis. It is clear from the Fig. 2 that maximum dry mycelial weight (240 mg) was obtained on Richard's solution and minimum (75 mg) on Elliot's solution. Among the natural media, lowest dry mycelial weight was obtained on Potato Dextrose (150 mg). Growth of the fungus on synthetic media were light cottony tangled but on natural media, it was medium to heavy cottony tangled.

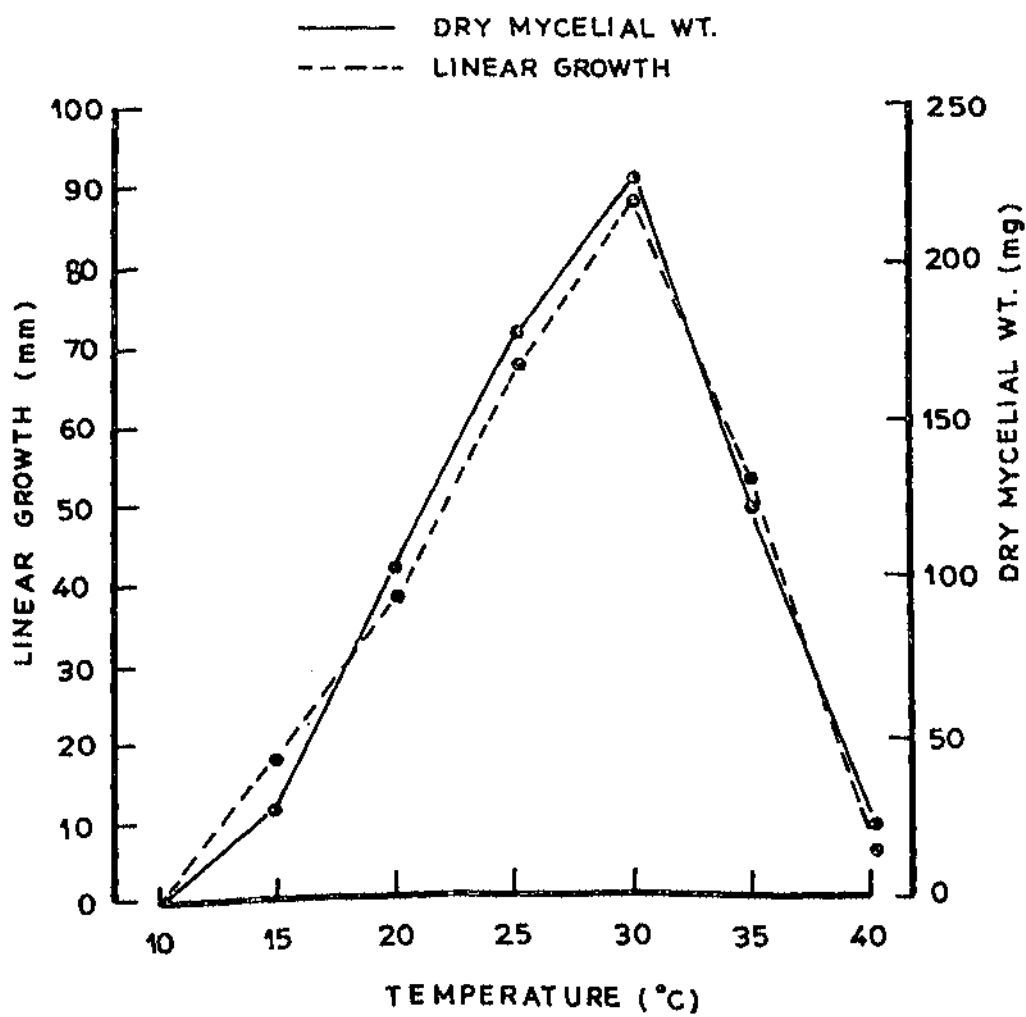


Fig.3. EFFECT OF TEMPERATURE ON GROWTH OF *Rhizoctonia bataticola*

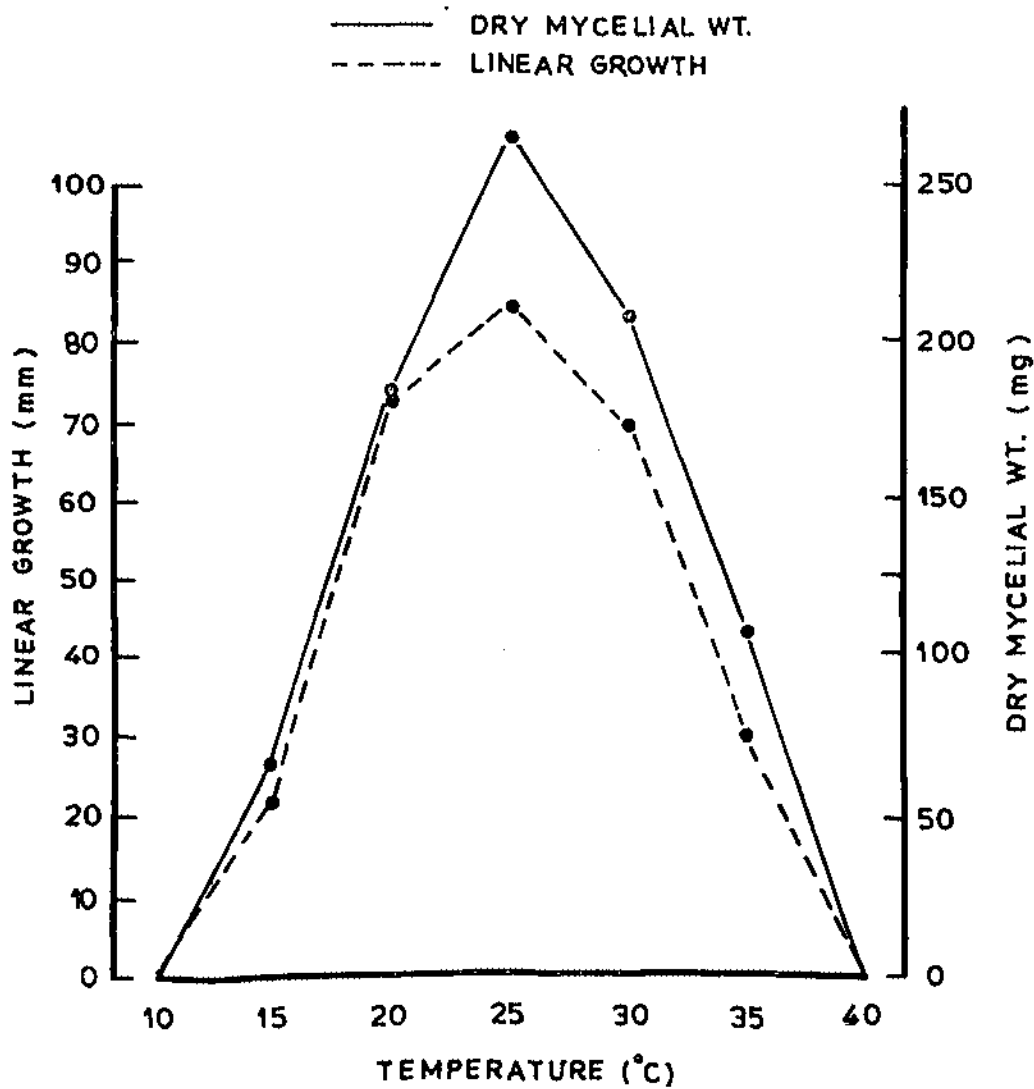


Fig. 4. EFFECT OF TEMPERATURE ON THE GROWTH OF
Fusarium oxysporum f.sp. ciceri

2.2 Effect of temperature on growth of *R. bataticola*:

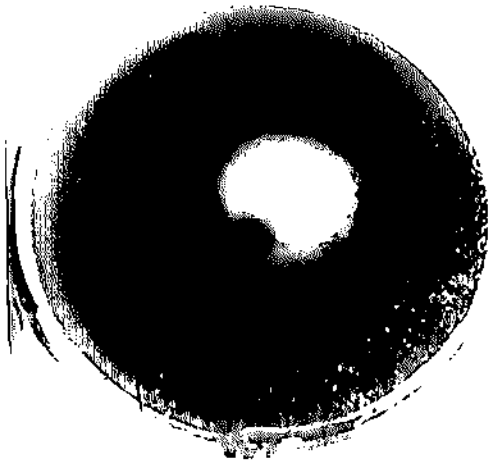
The linear growth and dry mycelial weight presented in Fig. 3 shows the growth behaviour after 60 hours and 7 days respectively. It is evident from the figure that *R. bataticola* grew best at 30°C (88 mm, 226 mg dry mycelial weight) followed by 25°C temperature (67 mm and 178 mg). The growth at 15°C, 20°C and 40°C was comparatively slow and started after 24 hours. There was no growth at 10°C in solid as well as liquid medium, however, at 40°C only 6 mm linear growth and 18 mg dry mycelial weight were recorded.

Effect of temperature on growth of *F. oxysporum* f.sp. *ciceri*:

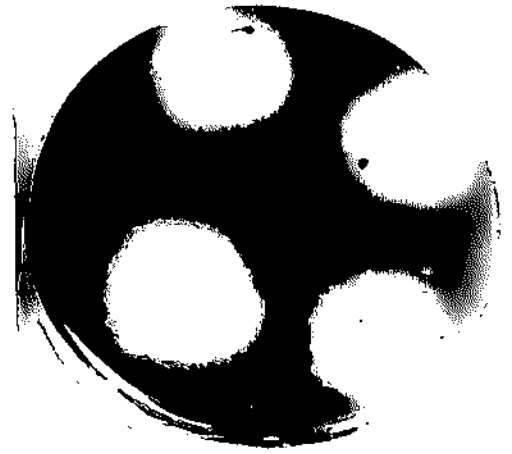
Linear growth and dry mycelial weight of *F. oxysporum* f.sp. *ciceri* grown at different temperature are presented in Fig. 4. It is clear that *F. oxysporum* f.sp. *ciceri* grew well at temperature 25°C (85 mm linear growth, 269 mg dry mycelial weight) followed at 20°C. The growth was not recorded at 10° and 40°C.

3.1 Effect of *F. oxysporum* f.sp. *ciceri* on the growth of *R. bataticola* in in-vitro:

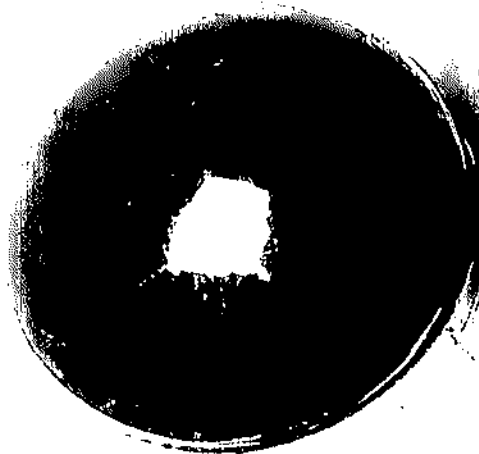
Data presented in Table 4 indicates linear growth after 48 hours of both the pathogens. It was noted that *R. bataticola* covered 44 mm while *F. oxysporum* f.sp. *ciceri* covered 22 mm in 90 mm Petriplate. When bits of both the pathogens were kept in the centre of the Petriplates together, the *R. bataticola* grew upto 31.6 mm while *F. oxysporum* f.sp. *ciceri* grew only upto 10.3 mm only. As it is clear from the photograph-I, *R. bataticola*



1



2



3

Plates- showing inhibition of F. oxysporum f.sp. ciceri by R. bataticola when placed at 1) side by side.
 2) R. bataticola in centre and F. oxysporum f.sp. ciceri on periphery.
 3) F. oxysporum f.sp. ciceri in centre and R. bataticola on periphery.

covered Fusarium colony from all the sides. In another set, R. bataticola was kept in the centre and F. oxysporum f.sp. ciceri on the periphery of the Petriplates. It was noted that average colony diameter of F. oxysporum f.sp. ciceri was 5.6 mm, while R. bataticola was 30.3 mm and covered the whole plate as shown in photograph-II. In third case when F. oxysporum f.sp. ciceri was kept in centre and R. bataticola on the periphery, average growth of R. bataticola was 21.6 mm while F. oxysporum f.sp. ciceri could grow only 8.3 mm diameter shown in photograph-III. These observations indicate that R. bataticola grew faster than F. oxysporum f.sp. ciceri and covered it completely.

Table 4: Growth behaviour of pathogens in 'in-vitro'.

Treatments	Linear growth after 48 hours (mm)	
	<u>R. bataticola</u>	<u>F. oxysporum</u> f.sp. <u>ciceri</u>
Both side by side	31.6	10.3
When <u>R. bataticola</u> in centre	30.3	5.6
When <u>F. oxysporum</u> f.sp. <u>ciceri</u> in centre	21.6	8.3
<u>R. bataticola</u> alone	44	-
<u>F. oxysporum</u> f.sp. <u>ciceri</u> alone	-	22

3.2 Effect of *F. oxysporum* f.sp. *ciceri* on growth of *R. bataticola* in 'in-vivo':

3.2.1 Water culture technique: One per cent mycelial suspension of both the pathogens (*R. bataticola* and *F. oxysporum* f.sp. *ciceri*) was kept in different ratios (*R. bataticola*; *F. oxysporum* f.sp. *ciceri*) viz. 10:90; 20:80; 30:70; 40:60; 50:50; 60:40; 70:30; 80:20; 90:10 and 100 per cent pure of both the pathogens along with control (simple sterilized distilled water).

Table 5: Interaction between *R. bataticola* and *F. oxysporum* f.sp. *ciceri*.

Pathogen	No. of lesions on roots	Mean shoot length (mm)	Mean root length (mm)
<u><i>R. bataticola</i> : <i>F. oxysporum</i> f.sp. <i>ciceri</i></u>			
10:90	0	85 ¹	84 ¹
20:80	3	100	78
30:70	3	103	91
40:60	4	108	95
50:50	4	107	95
60:40	5	110	97
70:30	7	113	100
80:20	7	115	102
90:10	7	115	105
<u><i>R. bataticola</i> (pure)</u>	7	116	109
<u><i>F. oxysporum</i> f.sp. <i>ciceri</i> (pure)</u>	0	81	70
Control	0	119	110

1. Average of ten replications.

Data presented in Table-5 indicates that when R. bataticola alone or above 70 per cent was present in suspension, produced an average of seven lesions on the root after 72 hours and yellowing started after 36 hours. On the other hand, when the ratios between R. bataticola and F. oxysporum f.sp. ciceri were reduced the lesions development also reduced accordingly. No apparent lesion was noticed, when R. bataticola and F. oxysporum f.sp. ciceri were reduced the lesions development also reduced accordingly. No apparent lesion was noticed, when R. bataticola and F. oxysporum f.sp. ciceri were mixed in the ratio 10 : 90. When R. bataticola was present alone or above 80 per cent, the reduction in shoot and root lengths were not marked, however, F. oxysporum f.sp. ciceri alone reduced the shoot and root length drastically (81 mm and 70 mm respectively). Not only this, but effect of F. oxysporum f.sp. ciceri on shoot and root lengths was clearly seen upto ratio 50:50 as shown in Table 5. These observations clearly showed that R. bataticola produced lesions while F. oxysporum f.sp. ciceri was responsible for reduction of shoot and root length when present together.

3.2.2 Pot interaction: Data presented in Table-6 showed that at 5 per cent inoculum level of R. bataticola, F. oxysporum f.sp. ciceri in the ratio (1:1), the disease incidence was 76.6 per cent, while F. oxysporum f.sp. ciceri alone, produced no apparent lesion on the root. F. oxysporum f.sp. ciceri either alone or in combination reduced the shoot and root length and it was 75.5 mm and 67 mm as compared to control 112 mm and 107 mm respectively.

There was not much difference between shoot and root length when R. bataticola was present alone and in control. In F. oxysporum f.sp. ciceri inoculated pots, shoot and root length much less than in combination of F. oxysporum f. sp. ciceri and R. bataticola as shown in Table-6.

Table 6: Interaction between R. bataticola and F. oxysporum f.sp. ciceri in 'in vivo' and dry root-rot development.

Treatment	Percent disease incidence	Percent final stand	Mean shoot length (mm)	Mean root length (mm)
<u>R. bataticola</u> (alone)	80 ¹	20	103 ¹	93 ¹
<u>F. oxysporum</u> f.sp. <u>ciceri</u> (alone)	-	100	61	53
<u>R. bataticola</u> + <u>F. oxysporum</u> f.sp. <u>ciceri</u> (50:50)	76.6	23.4	74.5	67
Control (uninoculated soil)	-	100	112	107

1. Average of four replications.

4.1 Effect of soil type on dry root-rot of chickpea caused by R. bataticola:

Sandy, sandy loam, clay loam and clay soils having different physio-chemical properties collected from different locations were sterilized and used for this study. Earthen pots (15 cm diameter) were filled with inoculated soil and 15 seed per pot, were sown. Five replications of each treatment were maintained. The data on dry root-rot incidence were noted after 25 days of sowing and are given in Table-7.

Table 7: Effect of soil type on the dry root-rot development of chickpea caused by R bataticola:

Soil type	Percent disease incidence	Percent final stand
Sandy soil	86.66 ¹ (66.14) ²	13.34
Sandy loam soil	76.60 (61.14)	23.40
Clay loam soil	68.30 (55.82)	31.70
Clay soil	61.60 (51.75)	38.40

C.D. at 5 per cent (5.47)

1. Average of five replications,
2. Figures in parenthesis are angular transformed value.

It was observed that different soil type behaved differently in dry root-rot development of chickpea. Table-7 depicts that sandy soil having 95 per cent, sand, 2.5 per cent silt and 2.5 per cent clay resulted 86.66 per cent root-rot incidence followed by sandy loam soil 76.60 per cent. There was significant difference between sandy and clay loam soil. Minimum dry root-rot was observed in clay soil having 61.60 per cent dry root-rot. It is clear from the Table-7 that when sand percentage was increased in soil, the per cent disease incidence also increased.

4.2 Effect of nitrogen and phosphorus on the development of dry root-rot of chickpea:

Incidence of dry root-rot was recorded in relation to various doses of nitrogen and phosphorus. Data presented in Table-8 indicates that highest disease incidence (88.3 percent) was recorded where soil was supplemented with 20 kg N/ha. It was further noted that when doses of nitrogen decreased, the disease incidences reduced accordingly.

On the other hand when doses of phosphorus were enhanced from 0 to 60 ^{kg}/P₂O₅/ha, the disease incidences were decreased. With the result, the lowest disease incidence 48.3 per cent was recorded in the soil where phosphorus was added at the rate of 60 kg P₂O₅/ha. The disease incidence at 15 kg P₂O₅/ha and 20 kg P₂O₅/ha level was not significantly different.

As shown in Table-8, in general it has been observed that with increasing doses of nitrogen in relation to constant phosphorus level disease incidence increased significantly. On the other hand with increased level of phosphorus at the constant level of nitrogen, disease incidences reduced significantly. An interesting feature obtained was that when recommended doses of nitrogen and phosphorus were applied, disease incidence was not statistically different, from the treatment, where no fertilizer was applied. When highest doses of both the fertilizers were applied, the disease incidence was less than highest dose of nitrogen alone.

Table 8: Effect of nitrogen and phosphorus on the development of dry root-rot of chickpea:

Treatment	Percent disease incidence			
	Phosphorus level			
	P ₀	P ₁	P ₂	P ₃
N ₀	58.3 ¹ (49.80) ²	55.0 (47.87)	51.6 (45.95)	48.3 (44.04)
N ₁	58.3 (49.80)	58.3 (49.80)	53.3 (46.91)	50.0 (45.00)
N ₂	68.3 (55.71)	63.3 (52.74)	56.6 (48.83)	56.6 (48.43)
N ₃	88.3 (68.66)	83.3 (65.95)	78.3 (62.28)	76.6 (61.14)

C.D. at 5 percent (1.94)

1. Average of three replications,
2. Figures in parenthesis are angular transformed values.

5.0 Screening of fungicides:

5.1 In in-vitro:

Richard's agar medium was poisoned with a fungicides at 1, 10, 50, 200, 500 ppm concentrations (formulation basis) after sterilization and were poured in sterilized Petriplates (90 mm diameter) with four replications of each concentration. Similarly, all the fungicides were used to poison the medium in the same fashion. These Petriplates were inoculated with 5 mm diameter disc of fungal inoculum and suitable control was maintained (with out fungicides). Radial growth was measured after 60 hours. Growth inhibition was calculated on the basis of control. Data were plotted on logarithmic probit scale and ED₅₀ of each fungicides were calculated. Data presented in Table-9 indicates that ED₅₀ value of carbendazim was less as

Table 9: Screening of fungicides in in-vitro against R. bataticola

Fungicides	<u>Growth inhibition(percent)</u>					Toxicity index	ED ₅₀
	<u>Concentration $\mu\text{g/ml}$</u>						
	1	10	50	200	500		
Copperoxy chloride	2.2	6.3	36	71.8	78.5	194.8	60.6
Mancozeb	2.2	20	47.4	80.0	94	243.6	51.0
Maneb	10.4	20.7	50	53.3	61.1	195.5	50.00
Thiram	11.1	17.1	73.3	77.4	94.4	273.4	45.5
Captan	9.66	16.6	58.5	70.4	88.8	243.9	52.0
Captafol	13.3	38.5	53.5	78.5	100	283.5	49.5
Quintozene	29.6	43.3	58.5	71.8	91.1	294.1	20.0
Emisan	18.6	32.6	64.4	80	100	295.6	27.5
Celdion-S	30.0	53.3	72.6	84	94.4	334.3	12.0
Carboxin	32.6	43.0	63.2	80.0	100	319.3	16.0
Carbendazim	49.2	84.4	100	100	100	433.6	1.0
Control	100	100	100	100	100	500	-

compared to others fungicides followed by celdion-S (12.0), carboxin (16.0) and quintozone (20.0). The maximum ED₅₀ value was recorded in case of copper oxychloride (60.6) followed by maneb, mancozeb, captan, 50, 51, 52 respectively. Data on toxicity index ~~are~~ presented in Table-9 which also supports the above observation. Highest toxicity index (433.6) was recorded in case of carbendazim followed by celdion-S and carboxin where toxicity index was 334.3 and 319.3 respectively. The lowest toxicity index was obtained in copperoxychloride (194.8).

5.2 Seed treatment:

Thiram, captafol, quintozone, emisan, celdion-S, carboxin, carbendazim + thiram were selected on the basis of the performance in laboratory study as well as general recommendation for the control of diseases caused by R. bataticola. Data presented in Table-10 reveals that all fungicides gave significantly better disease control as compared to control (untreated). However, carbendazim reduced disease incidence from 91.08 per cent (control) to 28.8 per cent followed by carbendazim + thiram. Where disease incidence was 35.5 per cent. Among the protectants, quintozone protected the seedlings better as compared to others. In general, performance of systemic fungicides were superior than protectants.

Table 10: Effect of seed treatment with fungicides on the incidence of dry root-rot of chickpea:

Sr.No.	Fungicides	Percent disease incidence	Percent final stand
1.	Thiram	56.55 ¹ (49.49) ²	43.55
2.	Captafol	66.60 (54.70)	33.40
3.	Quintozene	55.50 (48.18)	44.50
4.	Emisan	60.00 (50.77)	40.00
5.	Celdion-S	51.01 (45.61)	48.99
6.	Carboxin	46.60 (43.05)	53.40
7.	Carbendazim + thiram	35.55 (36.54)	64.45
8.	Carbendazim	28.80 (32.44)	71.20
9.	Control (without fungicide)	91.06 (68.53)	9.94

C.D. at 5 per cent (3.75)

1. Average of four replications,
2. Figures in parenthesis are angular transformed values.

5.3 Pre-sowing drenching of soil:

The same eight fungicides mentioned in seed treatment experiment were applied for pre-sowing drenching of inoculated soil at the rate of 0.2 per cent (W/V). Data presented in Table-11

indicates that pre-sowing drenching with carbendazim and carbendazim + thiram were showing 35.27 and 35.50 per cent dry root-rot as compared to 91.06 per cent in control followed by celdion-S, where disease incidence 42.2 per cent. Although performance of all the fungicide was not good as in seed treatment. It is clear from the table that fungicides protected seedling significantly better than control, when applied as pre-sowing drenching. Among the protectants thiram protected seedlings better as compared to others.

Table 11: Effect of soil drench with fungicides on the incidence of dry root-rot of chickpea:

Sr.No.	Fungicides	Percent disease incidence	Percent final stand
1.	Thiram	55.5 ¹ (41.18) ²	44.5
2.	Captafol	77.7 (61.91)	22.3
3.	Quintozene	64.4 (53.39)	35.6
4.	Emisan	62.1 (52.09)	37.9
5.	Celdion-S	42.2 (40.50)	57.8
6.	Carboxin	53.3 (46.89)	46.7
7.	Carbendazim+thiram	35.5 (36.57)	64.5
8.	Carbendazim	35.27 (36.51)	64.73
9.	Control (untreated)	91.06 (72.84)	8.94

C.D. at 5 per cent (4.47)

1. Average of four replications,
2. Figures in parenthesis are angular transformed values.

5.4 Seed treatment and soil drenching:

In this set of experiment, seeds were treated with eight fungicides separately as mentioned in seed treatment experiment and sown in 15 cm diameter earthen pots having inoculated soil. Seven days old seedlings were drenched with same dose of fungicides. Data on disease incidence and final stand presented in Table-12, depicts that when seeds were treated with carbendazim and seedlings drenched with the same dose of carbendazim exhibited 13.3 per cent as compared to control 86.6 per cent disease incidence followed by carbendazim + thiram (18.3 per cent). Among the protectants, thiram performed better where disease incidence was recorded 36.6 per cent.

Table 12: Effect of seed treatment and soil drenching on the incidence of dry root-rot of chickpea:

Sr.No.	Fungicides	Percent disease incidence	Percent final stand
1.	Thiram	36.6 ¹ (37.25) ²	63.4
2.	Captafol	53.3 (46.91)	46.7
3.	Quintozene	50.0 (45.00)	50.0
4.	Emisan	53.3 (46.92)	46.7
5.	Celdion-S	23.6 (28.85)	76.4
6.	Carboxin	31.6 (34.23)	68.4
7.	Carbendazim + thiram	18.3 (25.31)	81.7
8.	Carbendazim	13.3 (18.43)	86.7
9.	Control	86.6 (64.69)	13.4

C.D. at 5 per cent (5.53)

1. Average of four replications,

2. Figures in parenthesis are angular transformed values.

6.1 Effect of soil type on the efficacy of selective fungicides:

6.1.1 Seed treatment: Carbendazim and carbendazim + thiram were found effective among treatments. In seed treatments experiment, seed of chickpea (cv. C-235) treated with above mentioned fungicides and sown in inoculated soil having different textural characteristics. Data on disease incidence and final stand were recorded after 25 days of sowing.

Table 13: Effect of soil types on the efficacy of carbendazim and carbendazim + thiram as seed treatment to control dry root-rot of chickpea.

Soil type	Carbendazim		Carbendazim + thiram	
	Percent disease incidence	Percent final stand	Percent disease incidence	Percent final stand
1. Sandy soil	28.33 ¹ (32.19) ²	71.67	35.00 ¹ (36.26) ²	65.00
2. Sandy loam soil	31.66 (34.23)	68.34	41.66 (40.19)	58.34
3. Clay loam soil	35.00 (36.26)	65.00	43.37 (41.16)	56.63
4. Clay soil	41.66 (40.19)	58.34	50.00 (45.00)	50.00

C.D. at 5 per cent (4.95) (2.97)

1. Average of four replications,
2. Figures in parenthesis are angular transformed values.

Data presented in Table-13 reveal that carbendazim alone, when applied in sandy soil showing 28.33 per cent disease incidence, meaning thereby, it protected seedling 71.67 per cent. Similarly, on the clay soil the carbendazim protected 58.34 per cent. The differences between sandy and clay soil in relation to disease incidence were highly significant, although

the differences between sandy and sandy loam were not statistically significant. A similar trend was obtained when carbendazim + thiram treated seeds were sown. Although in this case, the mortality of seedlings ~~were~~ more as compared to carbendazim alone e.g. when carbendazim + thiram treated seeds were sown in sandy soil, 35 per cent disease incidence occurred and in the clay it was 50 per cent. In this case disease incidence differed between sandy and sandy loam, but between sandy and clay soil it was highly significant.

6.1.2 Pre-sowing soil drenching: Inoculated different types of soils were drenched with carbendazim and carbendazim + thiram at the rate of 0.2 per cent (W/V) after five days of incorporation of inoculum. Two days after this, untreated seeds of gram were sown. Data on disease incidence presented in Table-14 indicate that pre-drenching of fungicides in different types of soils also reduced the disease incidence to a great extent. The minimum disease incidence (31.66 per cent) was in sandy soil and maximum (46.60 per cent) in clay soil. Differences in the incidence of disease in sandy and clay soil were highly significant, while between sandy loam and clay loam, it was non-significant, when soils were drenched with carbendazim. Disease incidences between sandy and sandy loam soils were not statistically significant. Similar trends were noticed, when soils were drenched with carbendazim + thiram, where minimum disease incidence (40.00 per cent) was noticed in sandy soil and maximum (53.34 per cent) in clay soil.

Differences between the disease incidence between sandy soil and clay soil was highly significant, but between clay loam and clay was non-significant.

Table 14: Effect of soil types on the efficacy of carbendazim and carbendazim + thiram as soil drenching to control dry root-rot of chickpea:

Soil type	Carbendazim		Carbendazim + thiram	
	Percent disease incidence	Percent final stand	Percent disease incidence	Percent final stand
Sandy soil	31.66 ¹ (34.23) ²	68.34	40.00 ¹ (39.23) ²	60.00
Sandy loam soil	36.66 (37.25)	63.34	43.33 (41.16)	56.67
Clay loam soil	41.66 (40.19)	58.34	46.67 (43.08)	53.33
Clay soil	46.60 (43.06)	53.40	53.34 (46.91)	46.66

C.D. at 5 percent (6.13)

(4.39)

1. Average of four replications,

2. Figures in parenthesis are angular transformed values.

6.2 Effect of nitrogen and phosphorus on the efficacy of selective fungicides:

6.2.1 Seed treatment: Chickpea seeds were treated with carbendazim and carbendazim + thiram and sown in inoculated soil having various levels of nitrogen and phosphorus. Data on disease incidences are presented in Table-15, which indicate that at 0 kg N/ha disease incidence was 8.33 per cent and at highest levels of nitrogen and phosphorus (20 kg N and 60 kg P₂O₅/ha), 35.00 per cent disease incidence was recorded, meaning thereby, the heavy fertilization inactivated the efficacy of fungicides. Similarly when seeds were treated

Table 15: Effect of nitrogen and phosphorus on the efficacy of carbendazim and carbendazim + thiram in disease controlling ability (dry root-rot of chickpea) applied as seed treatment.

Treatment	Carbendazim		Carbendazim + thiram	
	Percent disease incidence	Percent final stand	Percent disease incidence	Percent final stand
N ₀ P ₀	8.33 ¹ (16.59) ²	91.67	18.34 ¹ (25.31) ²	81.66
N ₀ P ₁	6.66 (14.75)	93.34	21.67 (27.71)	78.33
N ₀ P ₂	3.33 (10.52)	96.67	18.34 (25.31)	81.66
N ₀ P ₃	3.33 (10.52)	96.67	16.67 (24.05)	83.33
N ₁ P ₀	10.00 (18.43)	90.00	21.67 (27.71)	78.33
N ₂ P ₀	25.00 (30.00)	75.00	46.67 (43.08)	53.33
N ₃ P ₀	45.00 (48.17)	55.00	61.67 (57.27)	38.33
N ₁ P ₁	11.66 (19.88)	88.34	23.34 (28.85)	76.66
N ₁ P ₂	8.33 (16.59)	91.67	21.67 (27.70)	78.33
N ₁ P ₃	6.66 (14.75)	93.34	20.00 (26.57)	80.00
N ₂ P ₁	28.33 (32.14)	71.67	41.66 (40.19)	58.34
N ₂ P ₂	31.06 (34.23)	68.94	40.00 (39.23)	60.00
N ₂ P ₃	25.00 (30.00)	75.00	38.34 (38.29)	61.66
N ₃ P ₁	45.00 (42.13)	55.00	60.00 (50.77)	40.00
N ₃ P ₂	41.66 (40.19)	58.34	56.67 (48.83)	43.33
N ₃ P ₃	35.00 (36.27)	65.00	53.34 (46.91)	46.66
			(2.67)	

C.D. at 5 per cent (3.93)

1. Average of four replications,
2. Figures in parenthesis are angular transformed values.

with carbendazim + thiram lowest (18.34 per cent) disease incidences were observed when no nitrogen and phosphorus were added. On the other hand, 53.3 per cent when heavy fertilizers doses (20 kg N and 60 kg P_2O_5 /ha) showed, inactivation of carbendazim alone and carbendazim + thiram. As interesting feature was noted that increased level of phosphorus expressed less disease incidence and therefore, inactivation of fungicides was less. Although the differences between phosphorus levels were significant. The disease incidence varied from 6.66 to 3.33 per cent, when phosphorus levels increased from 20 to 60 kg P_2O_5 /ha. A same trend was found when seeds were treated with carbendazim + thiram. Although disease incidences were slightly more, hence the inactivation of fungicide was comparatively more.

In case of nitrogen alone inactivation was more because at 10 kg N/ha, disease incidence was 10 per cent and at 20 kg/ha, it was 45 per cent, when seeds were treated with carbendazim and 21.67 per cent at 10 kg N/ha and 61.67 per cent at 20 kg N/ha, when seeds were treated with carbendazim + thiram. In general, it was noted that increased doses of nitrogen with constant level of phosphorus, disease incidence was increased in both the cases. While increased level of phosphorus with constant level of nitrogen, disease incidence decreased. It is now clear that nitrogen inactivated alone as well as in combination, while phosphorus level inactivated very less as shown in Table-15.

Soil drenching:

Pre-sowing drenching with carbendazim and carbendazim + thiram of inoculated soil having different levels of nitrogen and phosphorus in general exhibited more disease as compared to seed treatment experiment. Data presented in Table-16 depicts that at 0 level fertilizer disease incidence was 11.67 per cent and 20 per cent when soils were drenched with carbendazim and carbendazim + thiram respectively. Similarly at higher doses (20 kg N and 60 kg P_2O_5 /ha) 40 per cent and 60 per cent disease incidence was recorded. In this case also general trend was same as in case of seed treatment. Data in Table showed that ^{with} increased phosphorus levels in both the cases disease incidences decreased from 11.67 to 6.67 per cent and from 20.66 per cent to 16.66 per cent, when soils were drenched with carbendazim and carbendazim + thiram respectively. Similarly, increase in nitrogen level led to increase in the disease incidence in both the treatments. Similarly, increased levels of phosphorus alone or in combination with nitrogen reduced disease incidence level.

Table 16: Effect of nitrogen and phosphorus on the efficacy of carbendazim and carbendazim + thiram in disease controlling ability (dry root-rot of chickpea) applied as pre-sowing soil drenching

Treatment	Carbendazim		Carbendazim + thiram	
	Percent disease incidence	Percent final stand	Percent disease incidence	Percent final stand
N ₀ P ₀	11.67 ¹ (19.88) ²	88.33	20.00 ¹ (26.57) ²	80.00
N ₀ P ₁	10.00 (18.43)	90.00	21.66 (27.70)	78.34
N ₀ P ₂	8.23 (16.59)	91.66	18.33 (25.31)	81.67
N ₀ P ₃	6.67 (14.75)	93.33	16.66 (24.05)	83.34
N ₁ P ₀	15.00 (22.79)	85.00	31.66 (34.23)	68.34
N ₂ P ₀	28.34 (32.14)	71.66	48.33 (44.04)	51.67
N ₃ P ₀	48.38 (44.04)	51.62	71.10 (57.86)	29.90
N ₁ P ₁	15.00 (22.79)	85.00	25.00 (30.00)	75.00
N ₁ P ₂	13.34 (21.34)	86.66	23.30 (28.85)	76.70
N ₁ P ₃	10.00 (18.43)	90.00	20.00 (26.57)	80.00
N ₂ P ₁	30.00 (33.21)	70.00	46.60 (43.08)	53.40
N ₂ P ₂	33.34 (35.25)	66.67	43.33 (41.16)	56.67
N ₂ P ₃	26.67 (31.07)	73.33	40.00 (39.23)	60.00
N ₃ P ₁	48.34 (44.04)	51.66	70.00 (56.79)	30.00
N ₃ P ₂	45.00 (42.13)	55.00	61.06 (51.75)	38.04
N ₃ P ₃	40.00 (39.23)	60.00	60.00 (50.77)	40.00

C.D. at 5 per cent (2.84)

(2.42)

1. Average of four replications.

2. Figures in parenthesis are angular transformed values.

DISCUSSION

DISCUSSION

Dry root-rot of chickpea caused by R. bataticola has been found a severe threat to the cultivation of chickpea crop in Haryana. This disease has been characterized by yellowing of lower leaflets at the early stage, later on, the entire leaves/leaflets become dark brown and finally the plant is dried-off. When plant is uprooted one can find the dark brown lesions on roots and rootlets. In the initial stage of infection small minute brown lesion develops on root surface, which later on, turns dark brown and girdles the roots. These lesions are formed on the tip or on middle of the roots. In severe cases it also attacks the collar regions and the affected portion shrinks. This disease has been noted at seedling as well as at flowering stage. Damage at seedling stage is more conspicuous, which results in poor plant population in the field. On the other hand, the disease also occurs at flowering/maturity stage, but not so severe. At this stage a mixed infection of R. bataticola and F. oxysporum f.sp. ciceri were recorded. During survey it was noted that chickpea is damaged by this disease upto 35 per cent. Although severity varied from 12.5 to 35 per cent depending upon the locations of the fields. While working on dry root-rot of chickpea Nene et al. (1979) reported the dry root-rot at post flowering stage from Central and Southern India. They have also noted that severity of dry root-rot is insignificant in Northern India. Working on the same problem Tripathi and Sharma (1983) noted the dry root-rot in severe form at seedling as well as post flowering

stage in Southern part of Haryana. Present observations are in accordance with this observations that this disease also taken heavy toll in other parts of Haryana, where chickpea is cultivated as 'Barani' crop (without irrigation). Although severity varies from field to field. In some samples collected from field, F. oxysporum f.sp. ciceri was also isolated along with R. bataticola. In such cases an interesting feature was noted that plant remained stunting. R. bataticola has been found to be associated with dry root-rot upto 90 per cent in fields observations. Therefore, it has been established as a pathogen of this disease. Hence most of the studies have been made on this pathogen.

R. bataticola has been found to grew better on Richard's medium followed by Czapek's medium. This observation is in accordance with the studies made by Chattapadhyay and Bhattacharya (1966), Ghosh and Sen (1973) and Shanmugum and Govindaswamy (1973b). However, Rana (1980) has found the Sabouraud medium. Grover and Sakhuja (1981) and Singh and Chohan (1982) found that Czapek's medium was best for growth of R. bataticola. The differences between the nutritional requirement of R. bataticola as observed by several workers is attributed to physiological variations in isolates. This indicates that these isolates differ in their nutritional requirement.

Similarly F. oxysporum f.sp. ciceri grew well on Richard's medium. This observation gets positive support from Buldeo and Rane (1978) and Eswara and Chaudhary (1985). In this case also

differences in nutritional requirement may vary according to isolates.

R. bataticola exhibited maximum growth at 30°C, although it can grow in the range of 15 to 40°C. Similar observations were made by Dhingra and Sinclair (1972), Singh et al. (1973), Shokes et al. (1977) and Rana (1980) while working on isolates of R. bataticola. F. oxysporum f.sp. ciceri has shown maximum growth at 25°C but it can grow in the range of 15 to 35°C. While working on F. oxysporum f.sp. ciceri with different host, Several workers have reported 25°C as optimum temperature for growth and sporulation (Szatala, 1962; Micoso and Mays, 1977; Shukla and Bhargwa, 1977 and Chohan, 1963). Present observations showed that R. bataticola and F. oxysporum f.sp. ciceri grew better in the range of 25 to 30°C.

In present investigation, F. oxysporum f.sp. ciceri has been isolated from diseased roots alongwith R. bataticola, although F. oxysporum f.sp. ciceri is responsible for wilt (Nene et al., 1979). Critical field observations indicated that R. bataticola in association with F. oxysporum f.sp. ciceri caused stunting of plants alongwith dry root-rot lesions. Therefore, some experiments have been conducted to find out the synergistic and/or antagonistic effect in the two pathogens in disease syndrome.

In laboratory, it is noted that when R. bataticola and F. oxysporum f.sp. ciceri allowed to grow together, the former pathogen covered the Petriplates earlier than later. No doubt,

F. oxysporum f.sp. ciceri is a slow growing organism as compared to R. bataticola, even then R. bataticola covered around the F. oxysporum f.sp. ciceri and do not allow it to grow. This clearly indicates that it might be restricted the growth of F. oxysporum f.sp. ciceri in nature. In water culture experiment, it has been noted that presence of F. oxysporum f.sp. ciceri alone or in combination with R. bataticola (50:50) reduced the shoot and root length but in case of R. bataticola alone or in combination with F. oxysporum f.sp. ciceri (70:30), there was no reduction in shoot, root length and number of lesions. Working on these lines, Pieczarka and Abawi (1978) noted that F. solani f.sp. phaseoli applied with mixed inoculum of Pythium ultimum and Rhizoctonia solani reduced the shoot length of bean. Nene et al. (1979) have isolated F. oxysporum f.sp. ciceri from chickpea at seedling stage and noted that seedling mortality may not be due to F. oxysporum f.sp. ciceri alone at this stage. This observation confirms the present finding that F. oxysporum f.sp. ciceri did not produce any apparent lesion except reduction in shoot, root length and weakening of the plants. This pathogen may help R. bataticola in early mortality of young seedlings and stunting of growth by reducing shoot and root length. This finding has been confirmed in screen house experiment, where soil was inoculated with R. bataticola and F. oxysporum f.sp. ciceri alone or in combination (1:1). In case of R. bataticola

inoculated soil, clearcut dry root-rot lesions were obtained without any reduction in shoot and root length. On the other hand, in case of F. oxysporum f.sp. ciceri alone, no clearcut lesions were obtained upto 25 days except reduction in shoot, root length and poor growth of seedlings. In case of mixture, dry root-rot as well as stunting of plants were common features. The above experiment confirms the field observations.

R. bataticola is a soil borne pathogen, therefore, soil involvement, particularly soil texture and nutritional status, may play a major role in dry root-rot of chickpea. It is noted that soil having less clay, expressed maximum disease (86.66 per cent) as compared to soil having more clay content (61.60 per cent) meaning thereby sandy soil which have less clay is found ideal for development of the pathogen as compared to clay soil (15% sand; 29% silt and 56% clay). Philip et al. (1969) pointed out that more aerated soil was suitable for the growth of R. bataticola. Chattopadhyay and Mustafee (1977) explained that more disease in soil having more sand content may be due to coarse and thick walled mycelium of R. bataticola. Apart from this, more dry root-rot of chickpea caused by R. bataticola in sandy soil as compared to clay soil can also be explained that R. bataticola flourish well in less available soil moisture (Ghafter and Erwin, 1969; Livingston, 1945 and Rao and Mukerji, 1972). Since sandy soil have low moisture retention capacity, therefore, it can serve as better medium for the growth and development of R. bataticola as compared to clay soil having comparatively more water holding capacity.

Nutritional status of the soil has much effect on the host, pathogen and interaction in disease causation of soil borne pathogens. Since, soil was rich in potassium, therefore, nitrogen and phosphorus levels were tried to find out their effect on development of dry root-rot of chickpea in present investigation. Working on R. bataticola, Thakurji (1974) noted that increased level of nitrogen led to increase the disease caused by R. bataticola. Similarly, Thayer and William (1960); Sivaparkasam et al. (1975) and Ramaswami and Shanmugam (1976) noted that increased level of phosphorus reduced the incidence of disease caused by R. bataticola. Author's observations are in agreement with above workers. It is noted that at zero level fertilizer, disease incidence was 58.3 per cent. A further increase in nitrogen level at zero phosphorus level, disease incidence increased accordingly e.g. at 15 kg N/ha, disease incidence was 68.3 per cent, and at 20 kg N/ha disease incidence was recorded as 88.3 per cent. On the other hand, when phosphorus levels were increased and nitrogen levels were kept constant, disease incidence was decreased at 60 kg P_2O_5 /ha and it was 48.3 per cent. It is also noted that when nitrogen and phosphorus were increased simultaneously, the effect was dominated by phosphorus.

Screening of fungicides against given pathogen in laboratory condition is essential feature before recommendation of chemical against a particular disease. No doubt, several workers have screened fungicides against R. bataticola, the

casual organism of root diseases of several crop plants. Effectivity of quintozone, ceresan dry, carbendazim, benomyl, carboxin and thiophanate-methyl have been proved beyond doubt against isolates of R. bataticola (Masih/^{et al.}1970; Jain and Khare, 1972; Rath and Mishra, 1978; Malhan, 1980; Hooda and Grover, 1983 and Dharamvir, 1984). In present study, carbendazim and carboxin among the systemics, quintozone and emisan among the protectants were found highly effective. This observation is in accordance with the above findings. No doubt, the value of ED₅₀ was calculated by these workers were slightly different than author's observation. This may be due to the variations in the environmental condition and precision of handling the experiment. Nevertheless, these results project the same picture.

Eight fungicides were selected on the basis of their recommendation against R. bataticola as well as their performance in in-vitro test. It is noted here that carbendazim exhibited less disease as compared to other fungicides. Although performance of other fungicides was found significantly superior than control. In other words, it may be concluded that carbendazim protected seedlings maximum (71.2 per cent). These observations are in accordance with Verma and Vyas (1977), Mishra (1980), Basman and Prasad (1981) and Chattopadhyay et al. (1981). In literature, quintozone has been found effective for the control of R. bataticola (Mathur and Singh, 1973 and Shukla and Singh, 1973).

In present investigation, the performance of quintozone was noted superior among the protectants, however, as compared to systemic fungicides, it was not so effective. This may be due to prolonged effect of systemic as compared to protectants.

Pre-sowing drenching is another method to control soil borne diseases by reducing the initial inoculum level. Therefore, R. bataticola inoculated soil was drenched before sowing. The results indicated that carbendazim protected seedlings maximum (64.73 per cent) followed by carbendazim + thiram (64.5 per cent). Working on the pre-sowing drenching against Rhizoctonia causing root-rot disease, Shehata et al. (1982) and Suhag and Duhan (1983) found the similar results. These observations are confirming the present findings of effectivity of carbendazim for the control of R. bataticola.

To achieve the maximum protection of chickpea seedlings against R. bataticola, an experiment was conducted, where treated seeds were sown in inoculated soil and emerged seedlings were drenched with the same dose of fungicides after seven days of sowing. Carbendazim reduced the disease incidence to 13.3 per cent and gave maximum protection of seedlings 86.7 per cent followed by carbendazim + thiram (81.7 per cent). These results are in accordance with the observations made by Tripathi et al. (1977).

To find out the reason of differential behaviour of fungicides under laboratory and field conditions, following experiments have been conducted. In present case, carbendazim alone or in combination with thiram were taken for various trials. Their ED_{50} value was quite less as compared to field doses. Soil type and nutrition has been taken into consideration to correlate the effectivity of these treatments in controlling the dry root-rot of chickpea.

When treated seeds were sown in different types of soil, an interesting feature was noted that different types of soil contracted different amount of diseases. All conditions were maintained identical except soil treatment. Minimum disease control (expressed by percent disease incidence and final stand) was noted in clay soil, while maximum disease control was noted in sandy soil. The reason may be clay content of different soil type because increased clay percentage reduced the disease controlling potential of carbendazim as well as in combination with thiram. Khan (1980) has also explained the poor efficacy of pesticides in clay soil due to chemical nature of clay particles. He has noted that clay minerals are negatively charged while particles are positively charged, therefore, these particles are easily adsorbed on clay particles. Hence, they are less available against the pathogen.

Ilays et al. (1976) while working on R. bataticola noted that reduction in the efficacy of fungicides under soil environment might be due to different absorption capacity of different types of soil. Apart from these, water holding capacity of different types of soil can also be correlated in differential behaviour of the carbendazim and carbendazim + thiram. In pre-sowing drenching treatment similar results have been noted. The efficacy of carbendazim alone and in combination with thiram behaved differently in different types of soil. The reason for this is same as described earlier.

Nutritional status of soil has a major role to play in disease development as well as efficacy of fungicides. Carbendazim and carbendazim + thiram were applied as seed treatment and pre-sowing soil drenching in inoculated soil having different levels of nitrogen and phosphorus, it is noted that nitrogen levels reduced the efficacy of carbendazim alone or in combination with thiram, which is more pronounced in case of carbendazim + thiram. Various levels of phosphorus did not reduce or very less the efficacy of the fungicides. An interesting feature observed that nitrogen alone or in combination with phosphorous also behaved in the same fashion. Working on the same line Malhan (1980) reported that the efficacy of carbendazim and thiophanate-methyl was reduced by addition of nitrogen + phosphorus + potassium. These observations are in accordance with the present findings

although role of phosphorus in reducing the efficacy of carbendazim and carbendazim + thiram. The differences may be due to differences in host-pathogen combination. These experiments clearly indicate that the efficacy of fungicides can be influenced by several factors. Hence it can be concluded that a more application of required dose of fungicides against a disease can not serve the purpose without knowing the macro and micro environment of disease development.

SUMMARY

SUMMARY

Dry root-rot of chickpea (R. bataticola) has been noted as one of the severe root disease, which occurs at seedling as well as flowering stage. Survey conducted during 1983-84 crop season revealed that maximum dry root-rot (35 per cent) was found in Gangwa village and lowest root-rot (12.5 per cent) was recorded at Kaul (Kurukshetra). Isolations from diseased roots showed that 68 to 90 per cent disease was caused by R. bataticola. Association of F. oxysporum f.sp. ciceri was also noted in some samples. R. bataticola was found pathogenic which produced symptoms of dry root-rot, while F. oxysporum f.sp. ciceri did not produce any apparent lesion, but caused stunting of plants.

Out of eight synthetic and natural media^{used} tested for growth and development of R. bataticola as well as F. oxysporum f.sp. ciceri. It was noted that Richard's medium (synthetic) supported maximum linear as well as mycelial growth. The maximum growth and sporulation of R. bataticola and F. oxysporum f.sp. ciceri was obtained at 30°C and 25°C respectively.

Interaction between R. bataticola and F. oxysporum f.sp. ciceri revealed that R. bataticola grew faster than F. oxysporum f.sp. ciceri and covered the Fusarium colony from all the sides. Water culture technique indicated that R. bataticola alone or in combination with F. oxysporum f.sp. ciceri alone or in combination with R. bataticola (90:10) did not show any lesion,

development but certainly reduced the shoot and root length. Similar observation was noted in pot culture experiment, when R. bataticola mixed in soil caused 80 per cent dry root-rot, However, F. oxysporum f.sp. ciceri did not cause any lesion, except stunting of the plants. When inoculum of both the pathogens were mixed in soil (50:50) stunting and dry root-rot were noted.

Effect of soil type on disease development indicated that 86.66 per cent dry root-rot occurred in sandy soil whereas 61.60 per cent dry root-rot was noted in clay soil. The amount of disease varied according to sand and clay proportion.

Effect of nitrogen and phosphorus on dry root-rot of chickpea was studied and found that increased doses of nitrogen increased the dry root-rot incidence from 58.3 (at 0 kg N/ha) to 88.3 per cent (at 20 kg N/ha). Similarly, phosphorus doses reduced disease incidence from 58.3 per cent (at 0 kg P_2O_5 /ha) to 48.3 per cent (at 60 kg P_2O_5 /ha).

Laboratory screening of fungicides against R. bataticola revealed that out of eleven fungicides tested, carbendazim showed ED_{50} value one and toxicity index 433.6 while highest ED_{50} value 60.6 and lowest toxicity index 194.8 was of copperoxychloride. ED_{50} value of other fungicides varied from 12 to 52.

Out of eight fungicides tested for seed treatment for the control of dry root-rot of chickpea, carbendazim protected seedlings maximum (71.2 per cent) as compared to other fungicides in systemic group. However, quintozone was better among the protectants. Pre-sowing drenching of the fungicides indicated same pattern, however, the disease control was less as compared to seed treatment.

Seed treatment and post sowing drenching reduced dry root-rot of chickpea from 86.6 to 13.3 per cent in carbendazim treated plots. It seems that seed treatment followed by a drench was suitable for the control of this disease.

Efficacy of carbendazim was reduced in clay soil as compared to sandy soil in screen house experiment. The reduction in disease controlling ability of carbendazim alone and in combination with thiram can easily be correlated with the clay content when applied as seed treatment and pre-sowing drenching.

In another experiment it was noted that nitrogen level reduced the efficacy of carbendazim and carbendazim + thiram, when applied as seed treatment and soil drench. Similarly, phosphorus application has opposite effect.



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APPENDIX

APPENDIX - I

Constitutents of the media used

I. Asthana and Hawker's medium

- i. Glucose - 1.0 g
- ii. Potassium nitrate - 3.5 g
- iii. Potassium dihydrogen phosphate - 1.75 g
- iv. Magnesium sulphate - 0.75 g
- v. Agar-agar - 20 g
- vi. Distilled water - 1000 ml

II Czapek's medium

- i. Sodium nitrate - 3.0 g
- ii. Potassium phosphate - 1.0 g
- iii. Magnesium sulphate - 0.5 g
- iv. Ferrous sulphate - 0.01 g
- v. Potassium chloride - 0.5 g
- vi. Sucrose - 30 g
- vii. Agar-agar - 20 g
- viii. Distilled water - 1000 ml

III. Elliot's medium

- i. Potassium monobasic phosphate - 1.36 g
- ii. Sodium Carbonate - 1.06 g
- iii. Magnesium sulphate - 0.5 g
- iv. Dextrose - 5 g
- v. Asparagine - 1.0 g
- vi. Agar - agar - 20 g
- vii. Distilled water - 1000 ml

IV. Richard's solution

- i. Potassium nitrate - 10.0 g
- ii. Potassium monobasic phosphate - 5 g
- iii. Magnesium sulphate - 2.5 g
- iv. Ferrous chloride - 0.02 g
- v. Sucrose - 50 g
- vi. Distilled water - 1000 ml

V. Tochinai's medium

- i. Peptone - 10.0 g
- ii. Potassium dihydrogen phosphate - 0.5 g
- iii. Magnesium sulphate - 0.25 g.
- iv. Maltose - 20.0 g
- v. Agar-agar - 20 g
- vi. Distilled water - 1000 ml

VI. Malt extract medium

- i. Malt extract - 20 g
- ii. Dextrose - 20 g
- iii. Peptone - 1.0 g
- iv. Agar-agar - 20 g
- v. Distilled water - 1000 ml

VII Corn meal medium

- i. Corn meal - 20 g
- ii. Dextrose - 20 g
- iii. Agar-agar - 20 g
- iv. Distilled water - 100 ml

VIII Potato-dextrose medium

- i. Peeled and sliced potato - 250 g
- ii. Dextrose - 20 g
- iii. Agar-agar - 20 g
- iv. Distilled water - 1000 ml

IX Control

- i. Agar-agar - 20 g
- ii. Distilled water - 1000 ml

APPENDIX - II

Texture of different soil types used.

Textural name	Sand %	Silt %	Clay %
Sandy soil	95	2.5	2.5
Sandy loam	80.2	7.8	12
Clay loam soil	57.5	18.5	24
Clay soil	15	29	56

APPENDIX-III

List of chemicals tested

Common name/ Trade name	Chemical name	Formulation (%)	Manufacturer/ Supplier
Blitox-50	Copper oxychloride	50 W.P.	Tata Fison Ltd., Bombay
Emisan	Methoxy ethyl mercury chloride	6 % Hg	Bayer (India) Ltd.
Mancozeb (Methane M-45)	Manganese ethylene bisdithiocarbamate plus zinc	75 W.P.	Indofil Chemicals Ltd., India.
Maneb	Manganese ethylene bisdithiocarbamate	75 W.P.	Indofil Chemicals Ltd., India.
Thiram (Arasan, TMTD)	Tetramethyl-thiuramdisulphide	75 W.P.	B.I. du Pont de Nevours Inc., U.S.A.
Captan	N-trichloromethyl-thio-4-cyclohexane- 1,2-dicarboximide	50 W.P.	Chevron Chem. Co., (Ortho Div.), U.S.A.
Captafol (Difolatan)	Cis-N-(1,1,2,2-tetrachloroethylthio)- 4-cyclohexane-1,2-dicarboximide	80 W.P.	Chevron Chem. Co., (Ortho Div.), U.S.A.
Quintozone (Brassiccol, PCNB, Terraclor)	Pentachloro nitrobenzene	75 W.P.	Olin Mathieson Chem Corp., U.S.A.
Celdion-S	Not disclosed	TF 130 - 10%	Takeda Chemical Ind. Ltd., Chuo-Ku, Tokyo, Japan.

contd....

Common name/ Trade name	Chemical name	Formulation (%)	Manufacturer/ Supplier
Carboxin (Vitavax, DCMO)	2,3-dihydro-6-methyl-5-phenyl- carbamoyle-1,4-oxathilin	76 W.P.	Uniroyal Chem Inc., Connecticut, U.S.A.
Carbendazim (Bavistin, MBC)	Methyl benzimidazole-2-ylcarbamate	50 W.P.	Bedische Aniline & Soda Fabrik AG., W. Germany

